DESIGN DOCUMENT

Designing a Multi-Agent Research System

Collaborative intelligence for comprehensive market analysis

Five specialized agents. One evolving shared understanding.

System	Multi-Agent Market Research Assistant
Domain	Consulting and Market Intelligence
Design Focus	State, coordination, adaptive workflow, synthesis
Output	Evidence-based market analysis report

Prepared from the original Problem 3 assignment brief

Executive Summary

This design proposes a coordinated multi-agent research system for producing comprehensive market analysis reports. A research coordinator interprets the client question, activates the right specialist agents, manages dependencies, and decides when the evidence is mature enough for synthesis.

The design does not treat the agents as isolated report writers. Every agent reads from and contributes to a structured collaborative state. Findings can therefore be challenged, extended, connected, or converted into new research tasks. The final report is built from synthesis threads that preserve the relationships among evidence, quantitative analysis, market trends, and competitor behavior.

Core design principle The system creates value when one agent's work changes what another agent investigates or concludes.

Design Goals

- Create shared understanding instead of merely collecting separate agent outputs.
- Make dependencies and handoffs explicit so agents receive the context they need.
- Adapt the workflow to the question rather than running every agent in the same order.
- Support investigation loops when evidence is incomplete, contradictory, or low confidence.
- Begin synthesis only after coverage and quality conditions are satisfied.

1. System Architecture

The architecture uses a coordinator pattern around a collaborative state. The coordinator has visibility into all specialists and the shared state, while specialists communicate through structured contributions and events. This reduces uncontrolled agent-to-agent links and makes the research process observable.

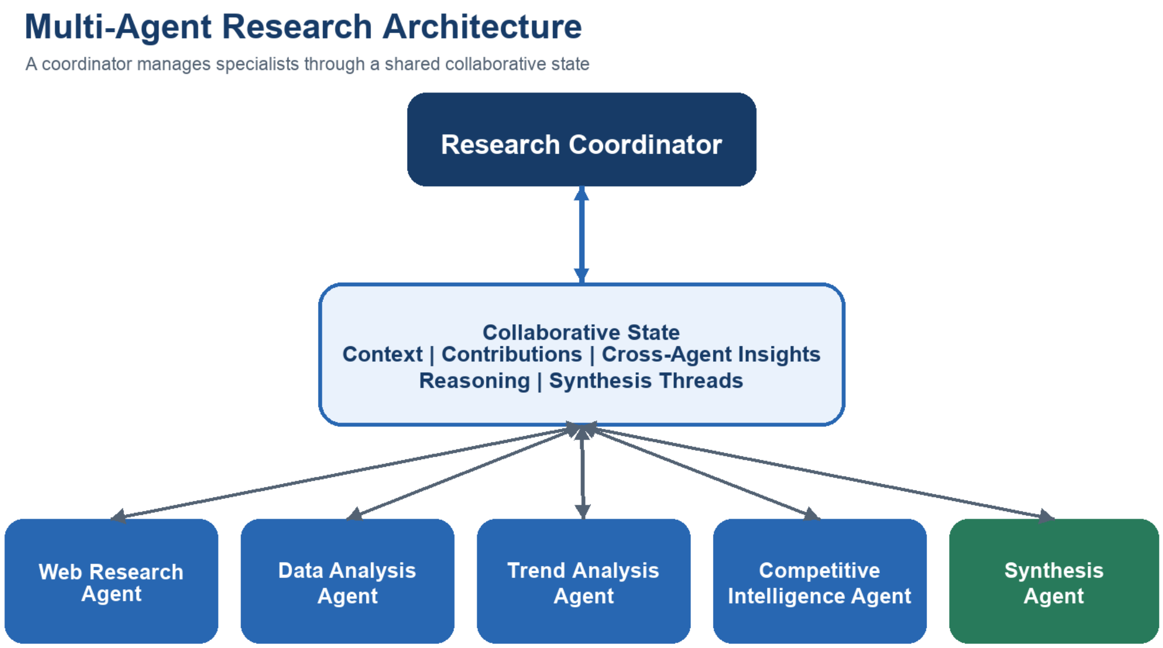

Figure 1. The coordinator selects and sequences specialists; the collaborative state carries shared knowledge.

Architectural Responsibilities

Component	Responsibility
Research Coordinator	Interprets the request, creates the plan, activates agents, resolves dependencies, and applies readiness gates.
Specialist Agents	Perform bounded research or analysis using the context and evidence relevant to their expertise.
Collaborative State	Stores research context, evidence, reasoning, links, unresolved questions, and synthesis threads.
Synthesis Agent	Builds an integrated report from validated threads rather than concatenating agent summaries.

2. Collaborative State Architecture

The shared state is organized into five layers. Each layer has a distinct purpose, owner expectations, and quality rules. This prevents the state from becoming a single unstructured container that every agent edits differently.

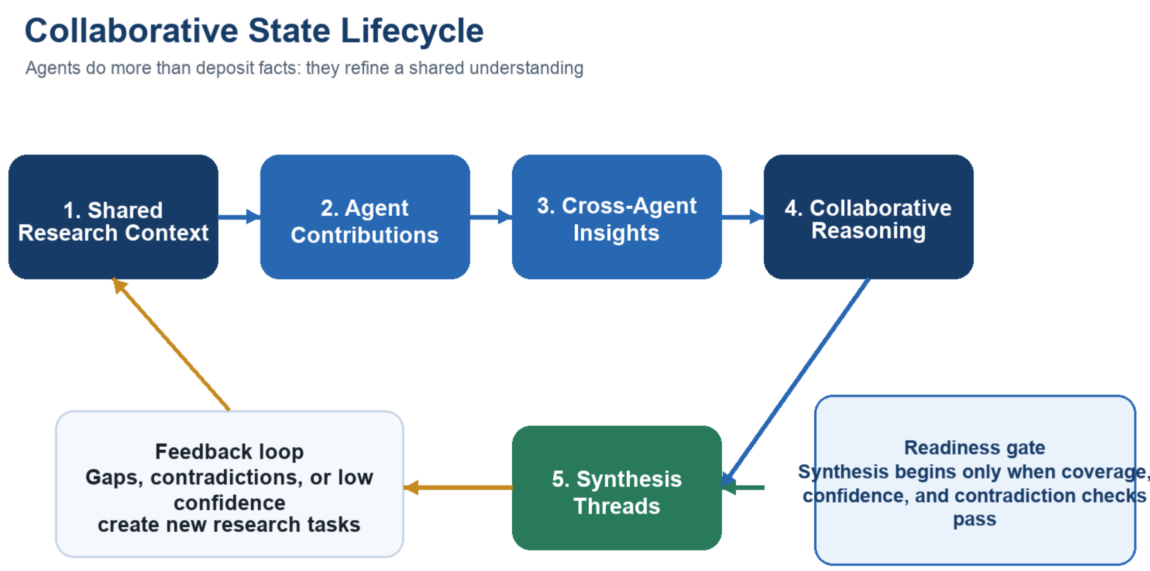

Figure 2. Contributions are converted into connected reasoning and synthesis threads, with feedback when gaps remain.

State Schema

State Layer	Representative Fields	Primary Writer	Purpose
Shared Research Context	research_question, client_objective, scope, geography, time_horizon, definitions, constraints, success_criteria	Coordinator	Stable framing used by every agent
Agent Contributions	agent_id, finding, evidence, source, confidence, assumptions, timestamp, related_question	Specialist agents	Traceable domain findings
Cross-Agent Insights	linked_findings, relationship_type, explanation, supporting_agents, confidence	Any agent or coordinator	Connections across domains
Collaborative Reasoning	hypotheses, confirmations, contradictions, open_questions, agreed_interpretations	Coordinator and specialists	Evolving shared understanding
Synthesis Threads	thread_title, claim, supporting_evidence, implications, risks, unresolved_items	Synthesis agent	Report-ready narratives

State Update Rules

1. Append evidence rather than silently replacing another agent's finding.
2. Attach a source and confidence level to every material claim.
3. Represent disagreement explicitly as a contradiction or competing hypothesis.
4. Link new insights to the findings that support them.
5. Promote content into synthesis threads only after evidence and dependency checks.

3. Agent Coordination Matrix

Each agent has a bounded role, but its work is intentionally connected to the others. The matrix below defines what each agent needs, contributes, responds to, and signals.

Agent	Needs	Contributes	Responds To	Signals
Web Research	Question, scope, keywords, unresolved evidence gaps	Articles, reports, facts, citations, source quality	New task; missing evidence; claim needing verification	Source found; weak source; conflicting external evidence
Data Analysis	Datasets, definitions, web findings, metrics to test	Calculations, comparisons, charts, anomalies, confidence	New dataset; quantitative claim; regional or temporal gap	Statistical pattern; outlier; insufficient data
Trend Analysis	Research evidence, data results, time horizon	Drivers, trajectories, scenarios, projections, uncertainty	Stable baseline evidence; material change signal	Emerging trend; turning point; projection risk
Competitive Intelligence	Market trends, company evidence, market metrics	Competitor moves, positioning, capability gaps, implications	Competitor event; trend with strategic impact	Strategic shift; market response; competitor contradiction
Synthesis	Validated findings, cross-agent insights, open issues, threads	Integrated narrative, implications, recommendations, report	Coverage threshold met; major contradictions resolved	Draft ready; evidence gap; final review required

Coordination Pattern

- The coordinator creates tasks with explicit inputs, expected outputs, and dependencies.
- Agents publish structured contributions to the shared state rather than sending informal messages.
- Important findings generate events that can activate another specialist.
- The coordinator tracks open questions, coverage, confidence, and contradictions.
- The synthesis agent remains downstream of evidence maturity, not simply elapsed time.

4. Dynamic Workflow Logic

The system first classifies the research request, then chooses an execution pattern. A narrow request can use a targeted path; independent evidence gathering can run in parallel; dependent reasoning is sequenced so later agents inherit validated context.

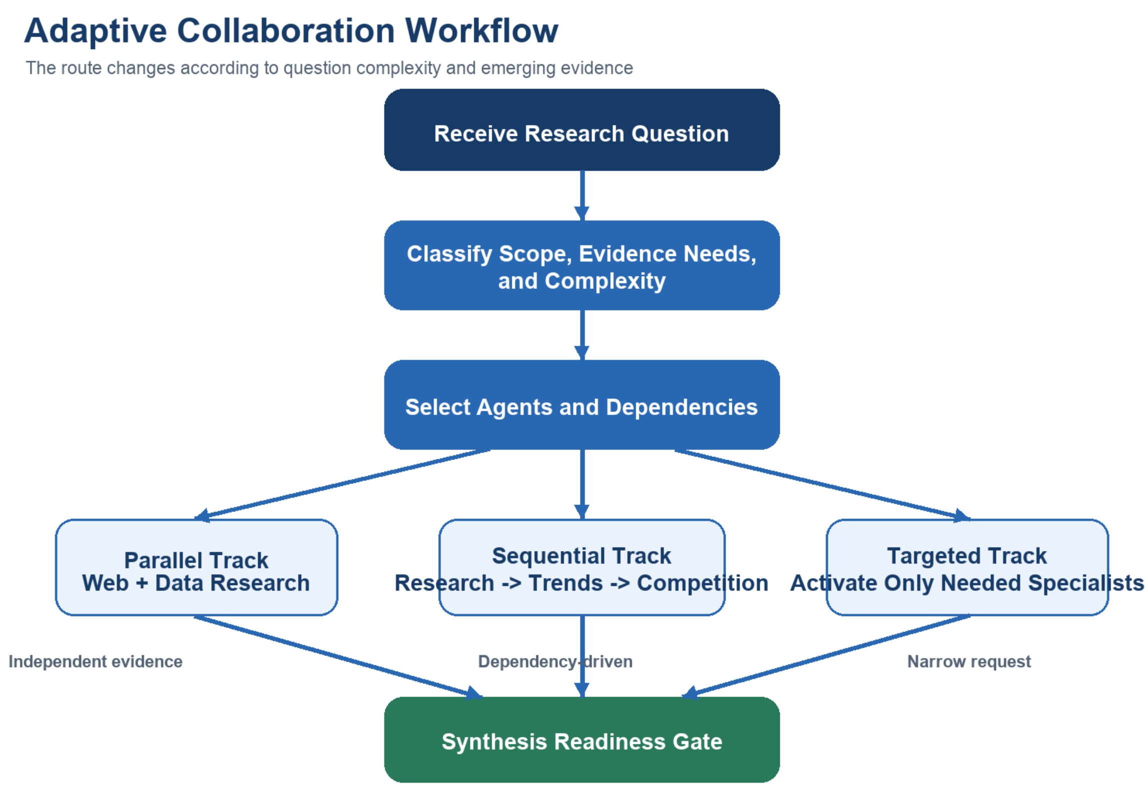

Figure 3. The coordinator selects parallel, sequential, or targeted collaboration and converges at a synthesis gate.

Agent Activation Framework

Question Type	Initial Activation	Dependency Logic
Factual or source-led question	Web Research	Data Analysis if quantitative validation is required
Market size or performance question	Web Research + Data Analysis in parallel	Trend Analysis after baseline validation
Future outlook question	Web Research + Data Analysis	Trend Analysis, then Competitive Intelligence
Competitor strategy question	Web Research + Competitive Intelligence	Data and Trend agents when market context is needed
Comprehensive market report	All evidence agents	Trend and Competitive agents build on research/data; Synthesis runs last

Parallel vs. Sequential Decision

- Run in parallel when agents can collect independent evidence from the same stable research context.
- Run sequentially when an agent's task definition depends on another agent's result.
- Use a hybrid workflow when broad evidence collection is parallel but interpretation requires ordered handoffs.
- Pause or backtrack when a critical input is absent, confidence is low, or findings materially conflict.

Synthesis Readiness Conditions

Readiness gate Begin synthesis when required domains are covered, major claims have evidence, confidence is acceptable, critical contradictions are resolved or disclosed, and no blocking research task remains.

5. Investigation, Backtracking, and Quality Control

Collaboration is not complete after the first pass. Agents may discover evidence gaps or contradictions that require additional work. The coordinator turns these discoveries into explicit follow-up tasks and preserves both the original finding and the resolution.

Condition	Detection	Backtracking Action
Missing evidence	A material claim has no reliable source	Web Research receives a targeted verification task
Quantitative inconsistency	Reported figures disagree across sources	Data Analysis reconciles definitions, dates, and units
Low-confidence projection	Trend conclusion depends on weak assumptions	Trend Analysis creates scenarios and records uncertainty
Competitive contradiction	Observed company action conflicts with stated strategy	Competitive Intelligence investigates behavior and timing
Narrative gap	A report thread lacks evidence or implication	Synthesis returns the thread to the relevant specialist

Completion Controls

- Coverage: every required report domain has at least one validated contribution.
- Traceability: important claims link to sources and originating agent contributions.
- Consistency: contradictions are resolved, bounded, or transparently reported.
- Confidence: uncertainty is recorded and reflected in conclusions.
- Coherence: synthesis threads connect evidence to implications rather than listing facts.

6. Worked Collaboration Example

Consider the research question: "How will the electric vehicle market evolve over the next three years, and where should a charging-network provider invest?" The example below shows how an isolated observation becomes a stronger strategic insight through collaboration.

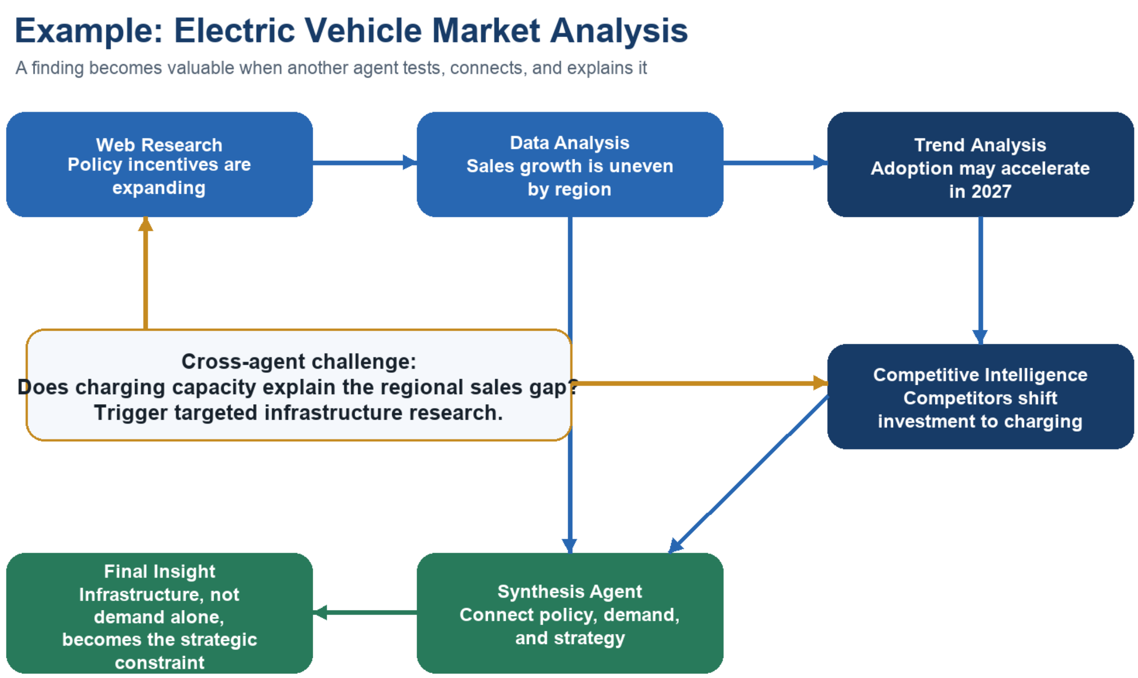

Figure 4. Cross-agent reasoning converts separate findings into a defensible strategic conclusion.

Example Reasoning Trace

1. Web Research identifies new incentives and infrastructure policy.
2. Data Analysis finds that sales growth varies sharply across regions.
3. Trend Analysis proposes that adoption will accelerate but records infrastructure as a limiting assumption.
4. Competitive Intelligence observes competitors redirecting investment toward charging capacity.
5. The coordinator creates a targeted question linking charging coverage to regional sales performance.
6. The synthesis agent forms a report thread: infrastructure availability may be the strategic constraint that determines where demand converts into adoption.

Why This Is Collaborative Intelligence

No single agent produced the final conclusion. Web evidence supplied policy context, quantitative analysis exposed regional variation, trend reasoning framed the future trajectory, and competitive analysis revealed how firms were responding. The shared state made these relationships visible, and the feedback loop prompted a new investigation. The final insight therefore emerged from connected reasoning rather than aggregation.

7. Report Assembly

The synthesis agent assembles the final deliverable from validated synthesis threads. Each section should preserve the relationship between evidence, interpretation, business implication, and uncertainty.

Report Section	Content
Executive Summary	Most important conclusions, implications, and recommended actions
Market Overview	Scope, size, segments, definitions, and baseline conditions
Data Insights	Quantitative findings, comparisons, anomalies, and limitations
Trends and Projections	Drivers, scenarios, time horizon, and uncertainty
Competitive Landscape	Competitor moves, positioning, responses, and strategic gaps
Integrated Implications	Cross-agent synthesis threads connecting market evidence to decisions
Conclusion	Priorities, risks, open questions, and next research steps

8. Collaboration Rationale

The proposed design satisfies the assignment's requirement for true collaborative intelligence in four ways.

Shared context creates alignment. All agents interpret the same objective, scope, definitions, and constraints.

Structured contributions create traceability. Findings retain evidence, confidence, assumptions, and ownership.

Cross-agent insights create new knowledge. The system records relationships that are not present in any isolated contribution.

Adaptive coordination improves relevance. Agents are activated, sequenced, or re-engaged according to the question and emerging findings.

Final design outcome The system produces a report whose conclusions are supported by connected evidence and collective reasoning, not a bundle of independent summaries.

9. Completion Criteria Mapping

Assignment Criterion	Where Addressed
Collaborative state enables intelligence beyond data sharing	Five-layer state architecture with explicit reasoning and synthesis layers
Coordination patterns show agents building on each other	Coordination matrix, dependencies, triggers, and event signals
Workflow adapts to research complexity	Parallel, sequential, targeted, and hybrid activation framework
Rationale explains emergent intelligence	Worked example and collaboration rationale